

● MEDIUM

● ALGEBRA

Linear functions

06

3 questions · ~5 min

Q1

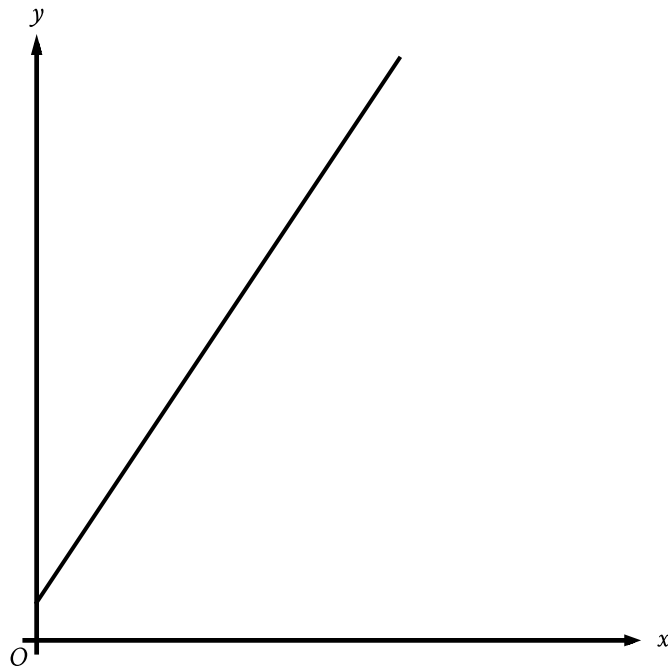
GRID-IN ALGEBRA

Brian saves $\frac{2}{5}$ of the \$ 215 he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in 9 weeks?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line begins on the y axis above the x axis.
- The line slants sharply up from left to right.

The graph represents the total charge, in dollars, by an electrician for x hours of work. The electrician charges a onetime fee plus an hourly rate. What is the best interpretation of the slope of the graph?

- A. The electrician's hourly rate
- B. The electrician's onetime fee
- C. The maximum amount that the electrician charges
- D. The total amount that the electrician charges

Q3

GRID-IN ALGEBRA

x	10	15	20	25
fx	82	137	192	247

The table shows four values of x and their corresponding values of fx . There is a linear relationship between x and fx that is defined by the equation $fx = mx - 28$, where m is a constant. What is the value of m ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.